

```
In [1]: import pandas as pd

df = pd.read_csv('allbacks.csv')
df
```

```
Out[1]:
```

	rownames	volume	area	weight	cover
0	1	885	382	800	hb
1	2	1016	468	950	hb
2	3	1125	387	1050	hb
3	4	239	371	350	hb
4	5	701	371	750	hb
5	6	641	367	600	hb
6	7	1228	396	1075	hb
7	8	412	0	250	pb
8	9	953	0	700	pb
9	10	929	0	650	pb
10	11	1492	0	975	pb
11	12	419	0	350	pb
12	13	1010	0	950	pb
13	14	595	0	425	pb
14	15	1034	0	725	pb

median cover type

```
In [2]: n = len(df)
sorted_covers = sorted(df['cover'])
median_idx = n // 2 # 0-based index of the middle value (n=15 is odd)
median_cover = sorted_covers[median_idx]

print("Sorted covers:", sorted_covers)
print("n =", n, "-> middle position (0-based index) =", median_idx)
print("Median cover type:", median_cover)
```

```
Sorted covers: ['hb', 'hb', 'hb', 'hb', 'hb', 'hb', 'hb', 'pb', 'pb', 'pb',
'pb', 'pb', 'pb', 'pb', 'pb']
n = 15 -> middle position (0-based index) = 7
Median cover type: pb
```

The median cover type is paperback and there are only 2 possible types so paperback is the most common.

mean weight of all books

```
In [3]: mean_weight_all = df['weight'].mean()
print("Mean weight (all books):", mean_weight_all)
```

Mean weight (all books): 706.6666666666666

The mean is about 706.7g, so on average each book weighs that much.

mean weight within each cover type

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In [4]: group_means = df.groupby('cover')['weight'].mean()
print(group_means)
```

```
cover
hb    796.428571
pb    628.125000
Name: weight, dtype: float64
```

The mean weight is very different between the two types (about 168g higher for hardcovers), which makes sense since hardcovers use heavier binding materials. For a bookstore this matters for shipping costs and storage weight, so mean weight by book type is a more useful reference than the overall mean.

verification

```
In [5]: hb_weights = df.loc[df['cover'] == 'hb', 'weight'].tolist()
hand_sum = sum(hb_weights)
hand_count = len(hb_weights)
hand_mean = hand_sum / hand_count

print("Hardcover weights:", hb_weights)
print(f"Sum = {hand_sum}, Count = {hand_count}, Mean = {hand_sum} / {hand_count}")
print("Matches pandas groupby result:", round(hand_mean, 2) == round(group_means['hb'], 2))
```

Hardcover weights: [800, 950, 1050, 350, 750, 600, 1075]

Sum = 5575, Count = 7, Mean = 5575 / 7 = 796.43

Matches pandas groupby result: True